



Responsible AI

Opportunity and Responsibility in the Era of Artificial Intelligence

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Responsible AI _ Agenda



Why

What

How

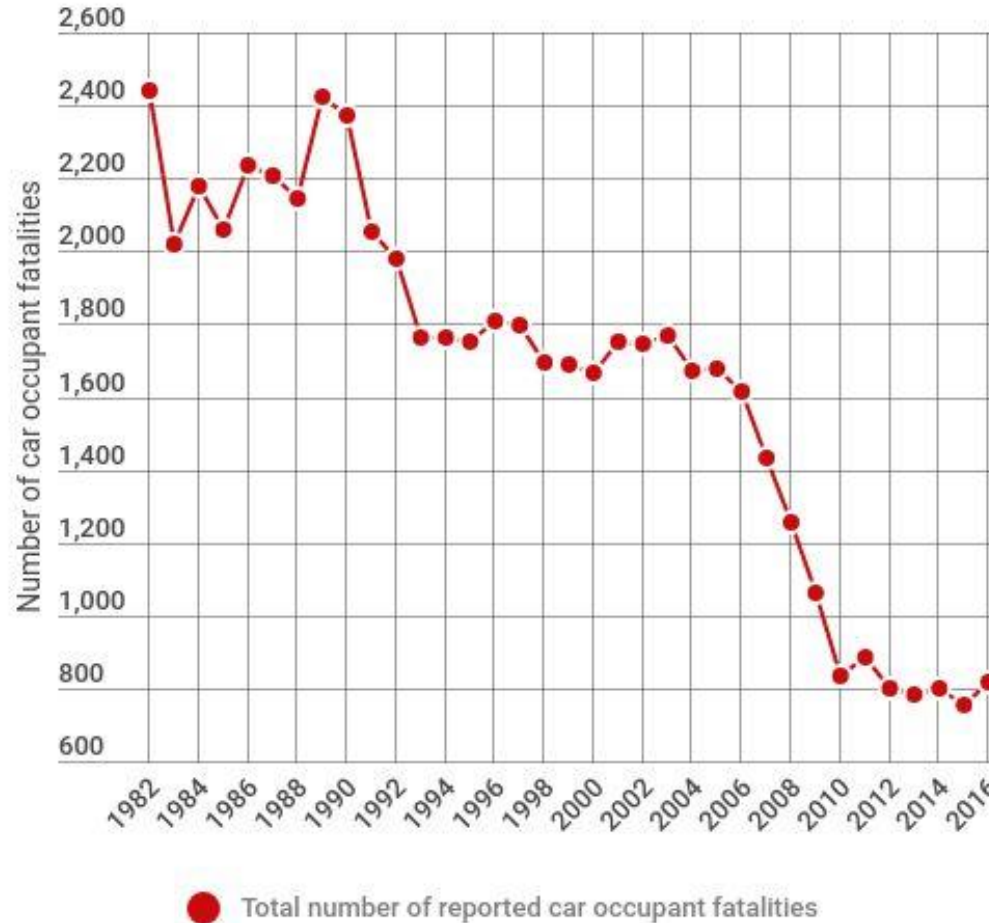


Unregulated vs Regulated Technologies

Seat Belts Save Lives

90.4%

SEAT BELT USE RATE IN 2021

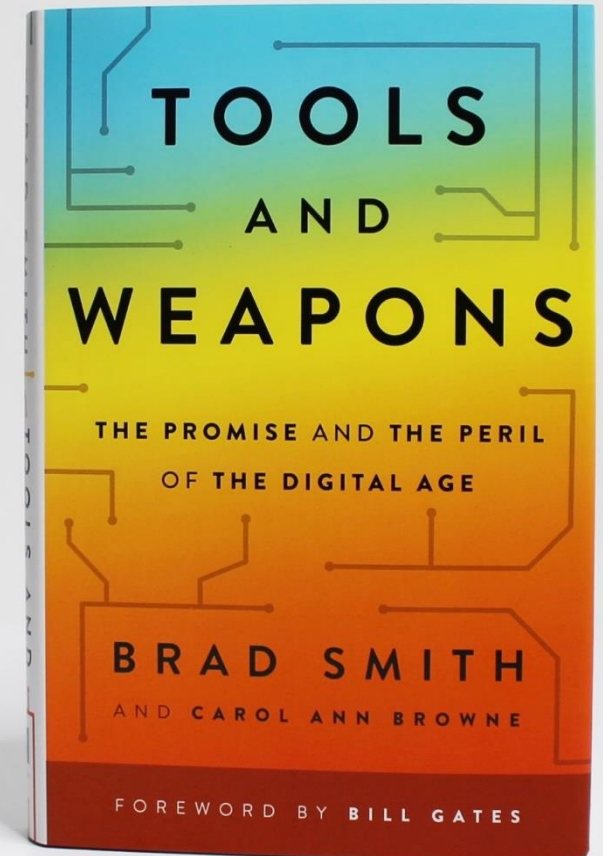


This information is based on National Statistics from the Department of Transport's 'Reported road casualties Great Britain, annual report: 2016'. Correct in January 2018.

Why responsible AI?

"The more powerful the tool, the greater the benefit or damage it can cause...Technology innovation is not going to slow down. The work to manage it needs to speed up."

Brad Smith
President and Chief Legal Officer, Microsoft



Challenges & Risks with AI

Challenge or Risk	Example
Bias can affect results	A loan-approval model discriminates by gender due to bias in the data with which it was trained
Errors may cause harm	An autonomous vehicle experiences a system failure and causes a collision
Data could be exposed	A medical diagnostic bot is trained using sensitive patient data, which is stored insecurely
Solutions may not work for everyone	A home automation assistant provides no audio output for visually impaired users
Users must trust a complex system	An AI-based financial tool makes investment recommendations - what are they based on?
Who's liable for AI-driven decisions?	An innocent person is convicted of a crime based on evidence from facial recognition – who's responsible?

Challenges & Risks with AI

Write 5 examples of challenges or risks with AI tools you have encountered recently:

- 1.
- 2.
- 3.
- 4.
- 5.

Today's debate

Anthropic releases Mythos-like AI model to the public two months after private rollout rocked Wall Street

- Anthropic announced **Claude Fable 5**, a Mythos-class model that will be available to its enterprise customers and paid subscribers.
- The company unveiled Mythos in April and has limited the rollout because of its advanced cybersecurity capabilities.
- Anthropic said Claude Fable 5's broad release is possible because of new safeguards that block responses in specific high-risk areas.

New

Fable 5: Our most capable model yet

Our newest model tackles your biggest challenges with fewer check-ins needed.

Included in your plan limits until Jun 22

Fable takes 2× the usage of Opus.

Switch models when a message is flagged



When safety measures flag a message, automatically switch to a different model to keep chatting. When off, your chat will pause instead.

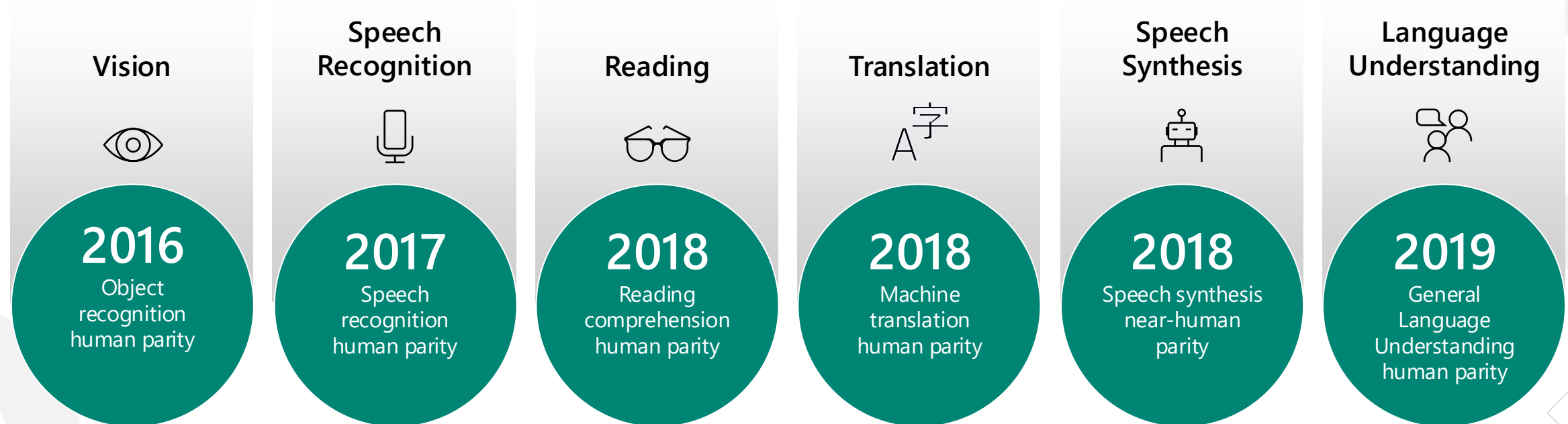
[Learn more](#)

Try Fable 5

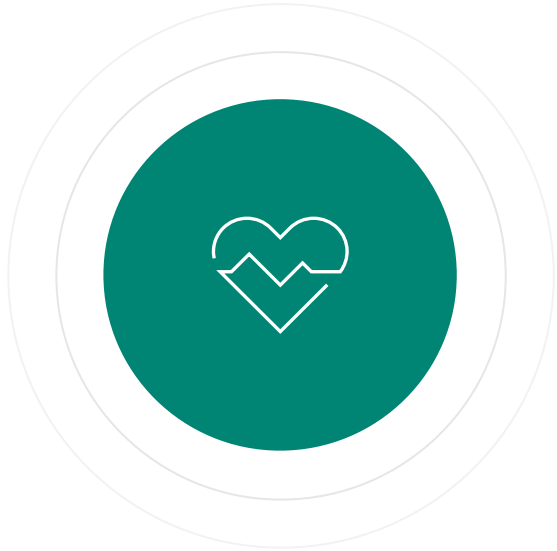


Why responsible AI?

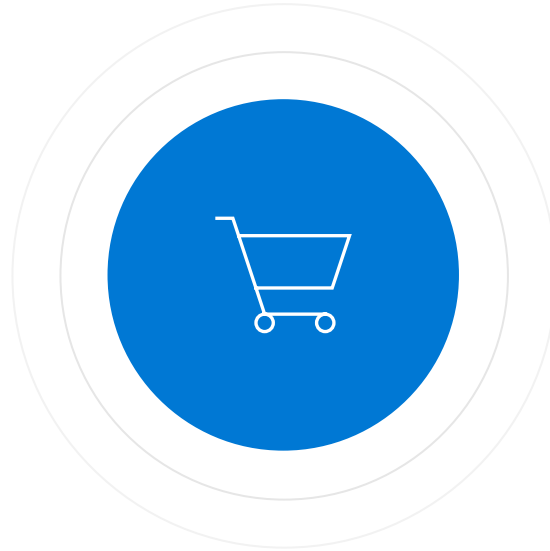
Advancements in AI are different than other technologies because of the **pace of innovation**, and its **proximity to human** intelligence – impacting us at a personal and societal level.



The Opportunities with AI



Healthcare



Retail



Financial Services



Manufacturing

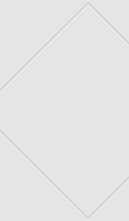
Agenda _Responsible AI



Why

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Microsoft AI Principles

The bottom of the slide features several abstract geometric shapes in various shades of teal and light teal. These include several elongated, rounded rectangles and one circle, all arranged in a dynamic, overlapping pattern that suggests movement and modern design.

Microsoft's AI principles



Fairness



Reliability
& Safety



Privacy &
Security



Inclusiveness



Transparency



Accountability

Microsoft's AI principles



Fairness



Reliability
& Safety



Privacy &
Security



Inclusiveness

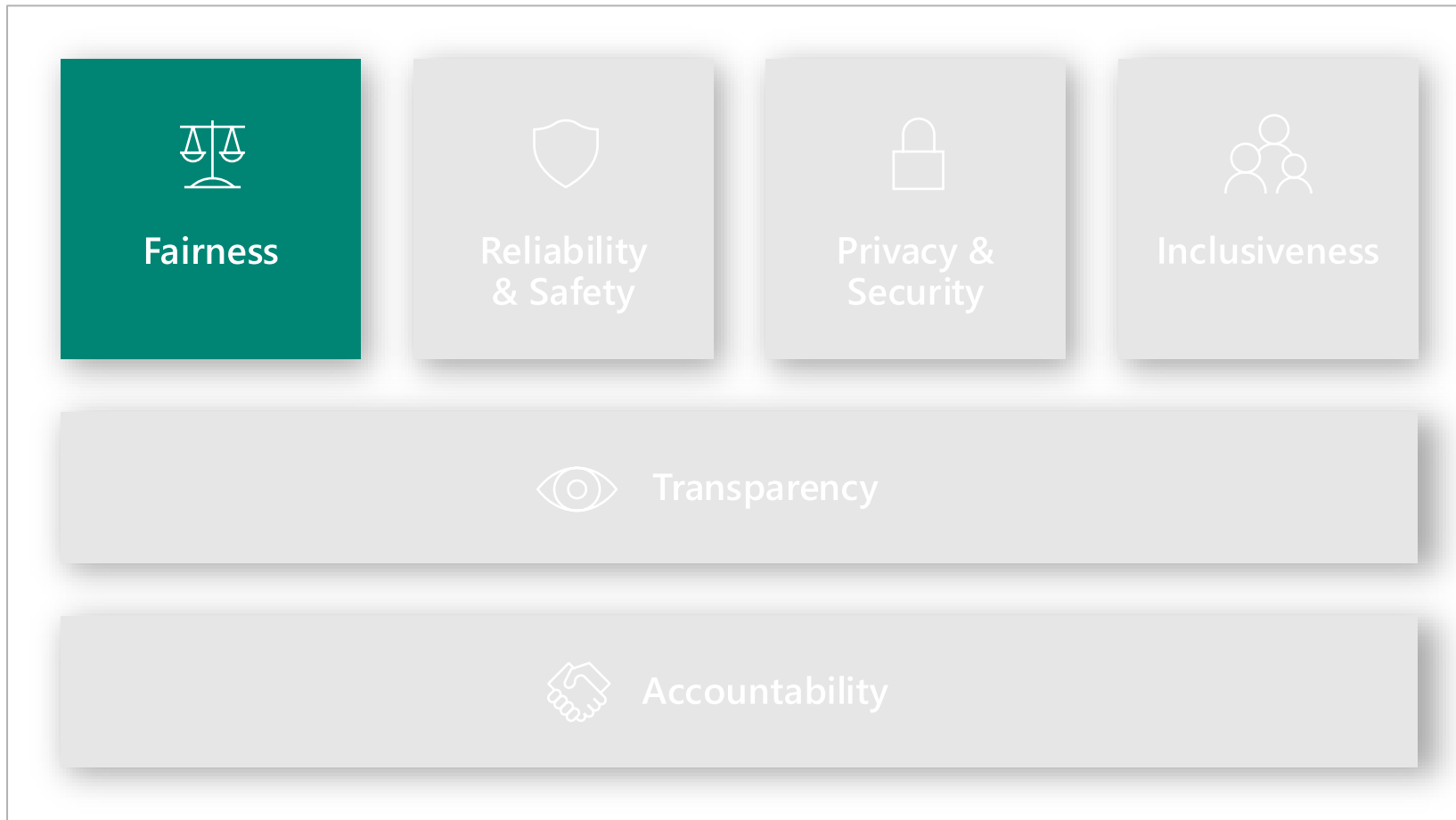


Transparency



Accountability

Microsoft's AI principles



Fairness

MENU ▾

nature

NEWS • 24 OCTOBER 2019

Millions of black people affected by racial bias in health-care algorithms

Study reveals rampant racism in decision-making software used by US hospitals – and highlights ways to correct it.

Heidi Ledford



...likely than equally ill white people to be referred to programmes that provide more personalized care. Credit: Ed Kashi/eyevine

Fairness

Is there gender bias in lending?

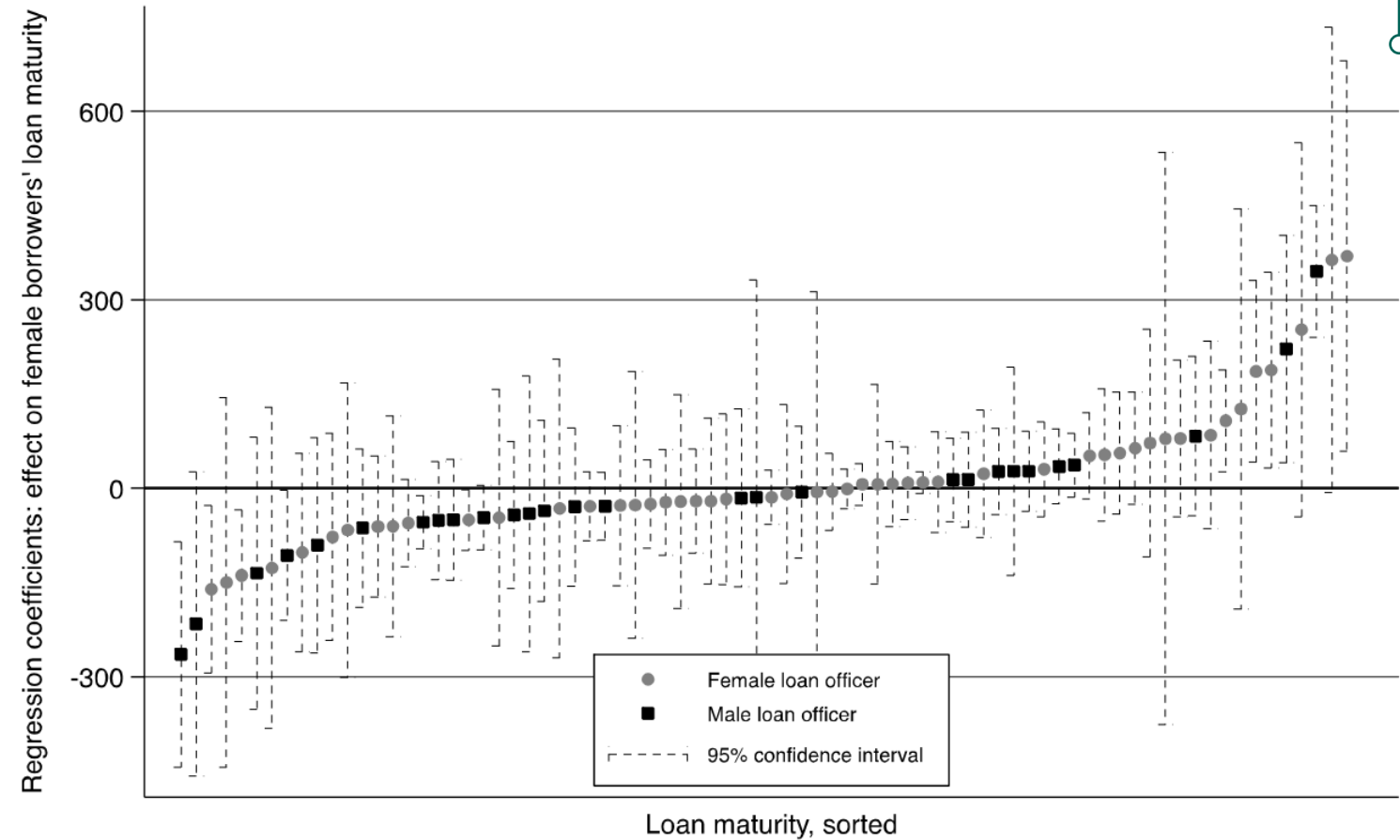
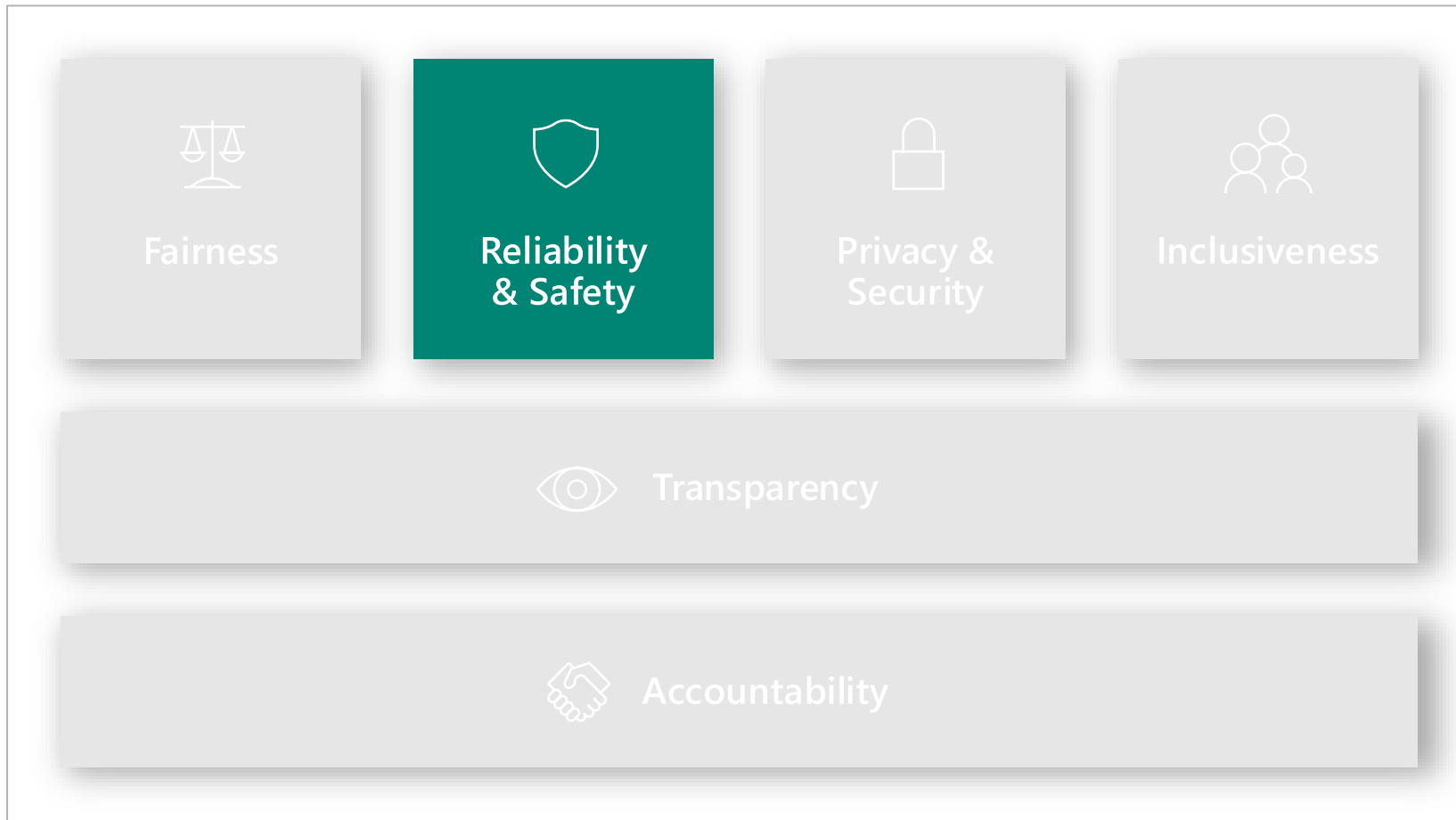


Figure 3. The figure shows the distribution of the bias on loan maturity by officer gender. Each coefficient represents an estimate of the number of extra days of loan maturity an individual officer approves for female versus male borrowers.

Source: SSRN, Sex and Credit: Is there gender bias in lending?, Beck, Behr & Madestom, April 2017

Microsoft's AI principles



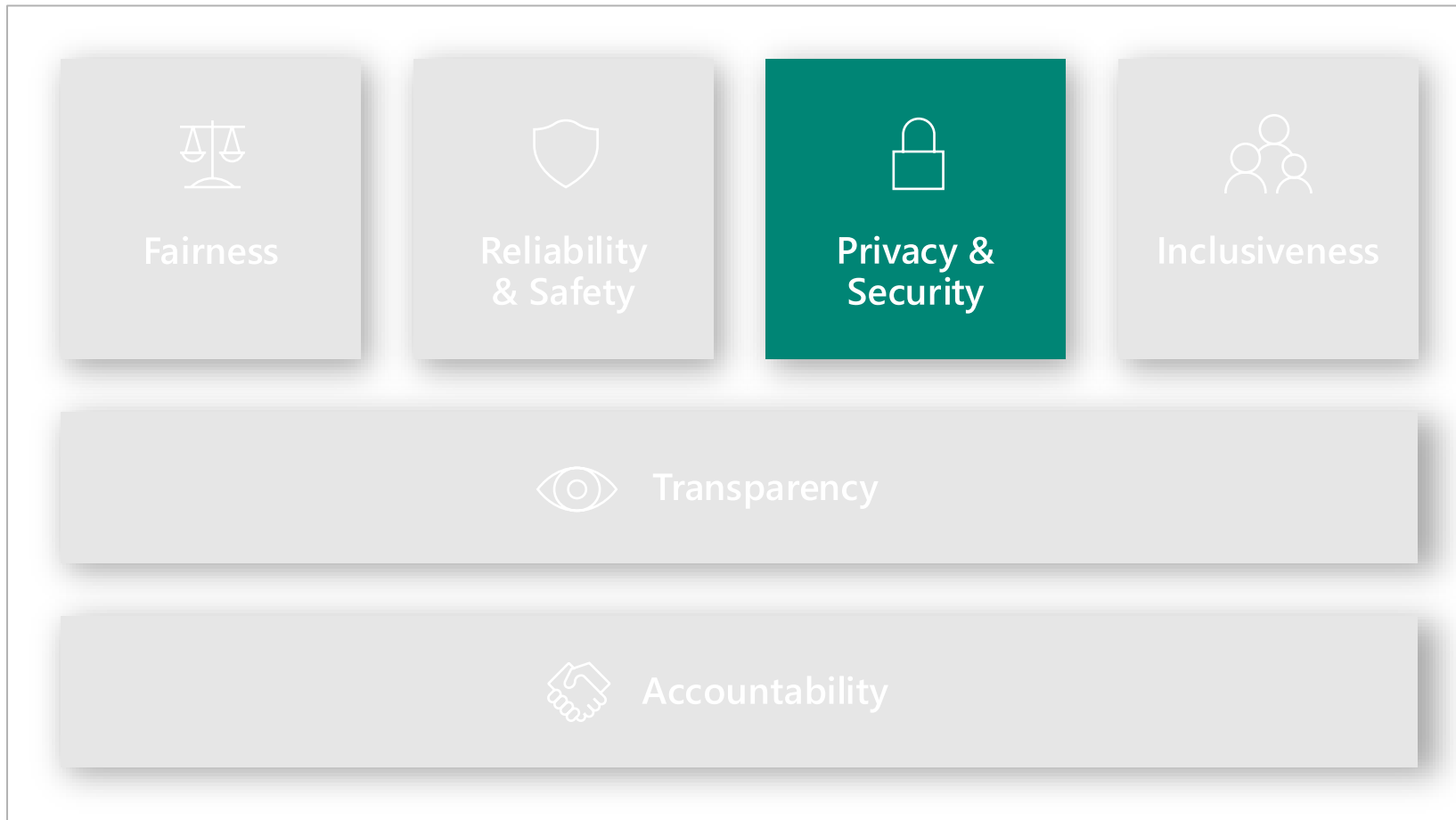
Reliability & Safety

Miles & years needed for to demonstrate autonomous vehicle reliability

Benchmark Failure Rate				
Statistical Question	How many miles (years*) would autonomous vehicles have to be driven...	(A) 1.09 fatalities per 100 million miles?	(B) 77 reported injuries per 100 million miles?	(C) 190 reported crashes per 100 million miles?
	(1) Without failure to demonstrate with 95% confidence their failure	275 million miles (12.5 years)	3.9 million miles (2 months)	1.6 million miles (1 month)
	(2) To demonstrate 95% confidence their failure to within 20% of the true rate of...	8.8 billion miles (400 years)	125 million miles (5.7 years)	51 million miles (2.3 years)
	(3) To demonstrate with 95% confidence and 80% power that their failure rate is 20% better than the human driver failure rate of...	11 billion miles (500 years)	161 million miles (7.3 years)	65 million miles (3 years)
*We assess the time it would take to complete the requisite miles with a fleet of 100 autonomous vehicles (larger than any known existing fleet) driving 24 hours a day, 365 days a year, at an average speed of 25 miles per hour.				

Source: Rand Corp. *Driving to Safety*; Kara & Paddock

Microsoft's AI principles



Privacy & Security



Privacy & Security



news

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Company suspends use of mall directory cameras running facial recognition software

Cadillac Fairview had been testing the technology since June, but didn't tell mall patrons

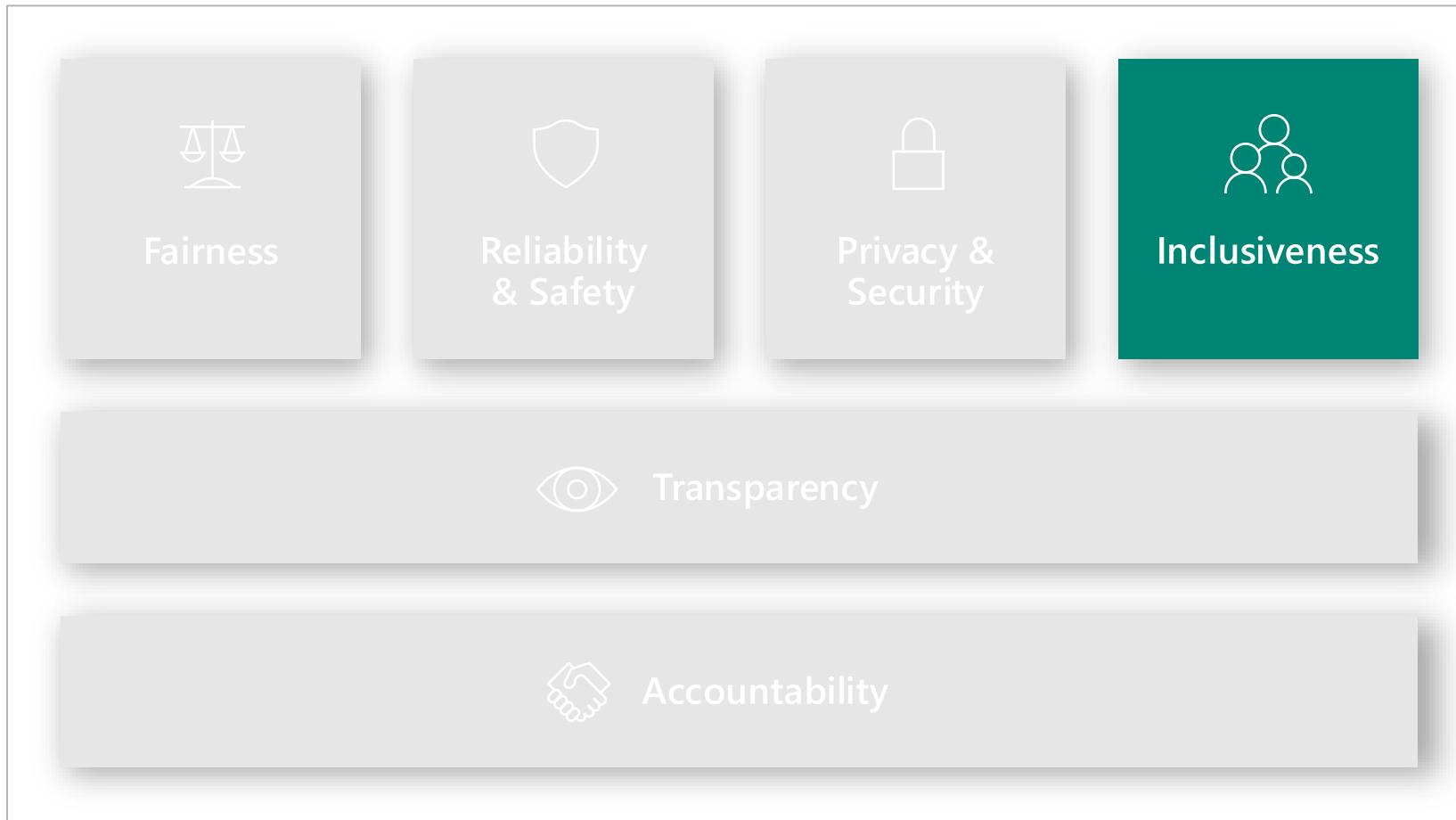


Anis Heydari · CBC News · Posted: Aug 04, 2018 6:08 PM MT | Last Updated: August 4, 2018



This mall directory at Cadillac Fairview's Chinook Centre in Calgary has a camera embedded within it, as circled in red on the left. (Anis Heydari/CBC)

Microsoft's AI principles



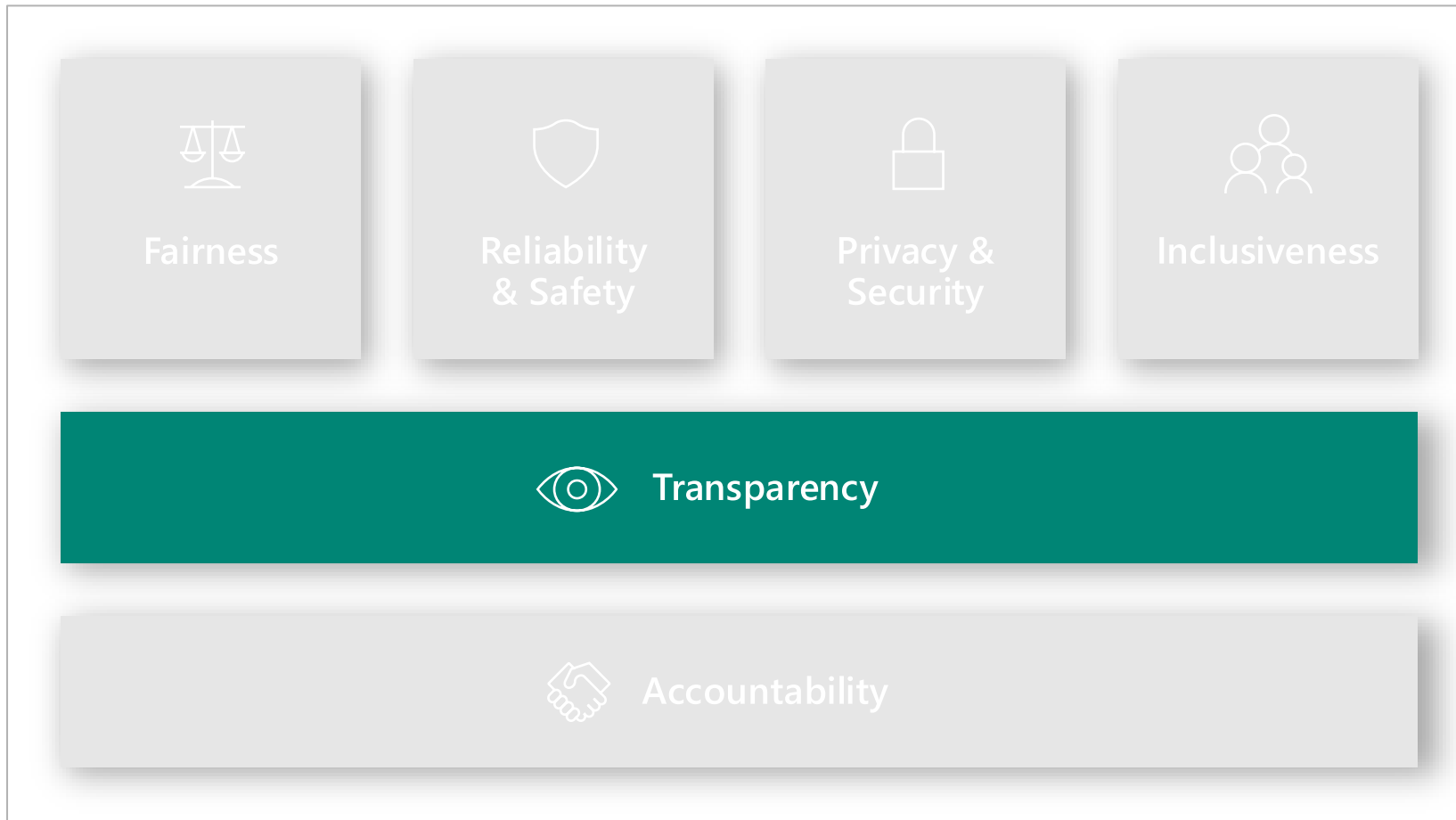
Inclusiveness



K. W. v. Armstrong plaintiff Christie Mathwig

https://www.acluidaho.org/sites/default/files/field_documents/class_action_complaint.pdf

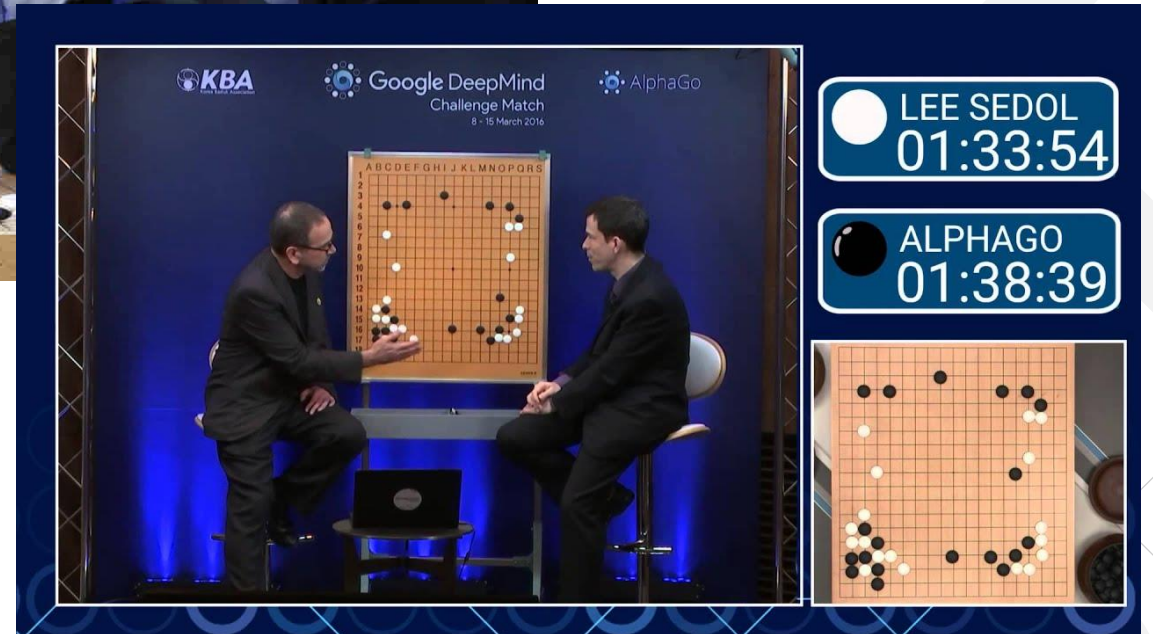
Microsoft's AI principles



Transparency



Move 37



Transparency

THE VERGE

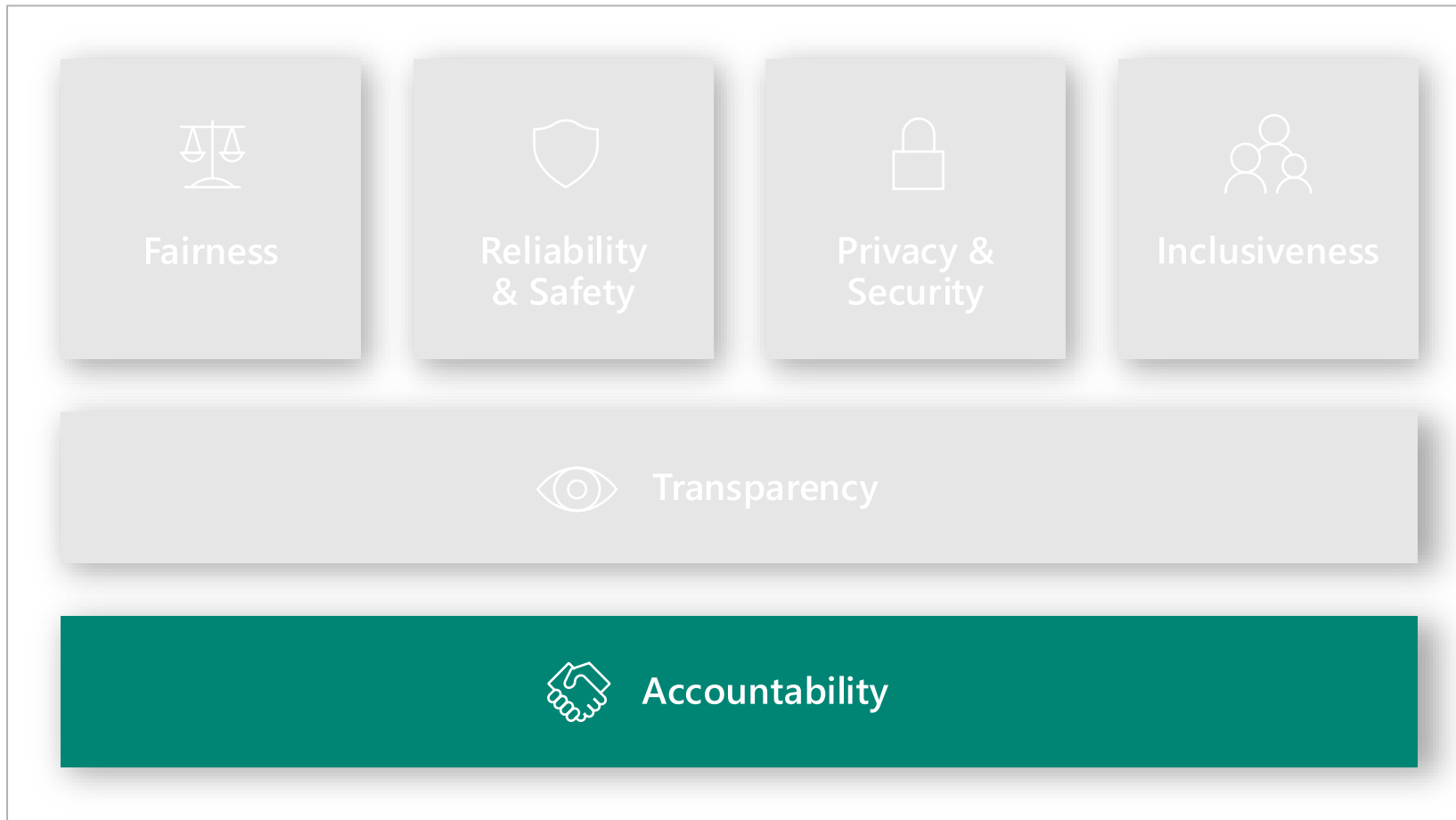
POLICY & LAW

PALANTIR HAS SECRETLY BEEN USING NEW ORLEANS TO TEST ITS PREDICTIVE POLICING TECHNOLOGY

Palantir deployed a predictive policing system in New Orleans that even city council members don't know about

By [Ali Winston](#) | Feb 27, 2018, 3:25pm EST

Microsoft's AI principles



Accountability

MIT Technology Review

Ethical Tech / AI Ethics

When algorithms mess up, the nearest human gets the blame

A look at historical case studies shows us how we handle the liability of automated systems.

by Karen Hao

May 28, 2019



Accountability

UK Official Says It's Too Expensive to Delete All the Mugshots of Innocent People in Police Databases



Sidney Fussell

4/19/18 2:30pm • Filed to: SURVEILLANCE ✓



16



4

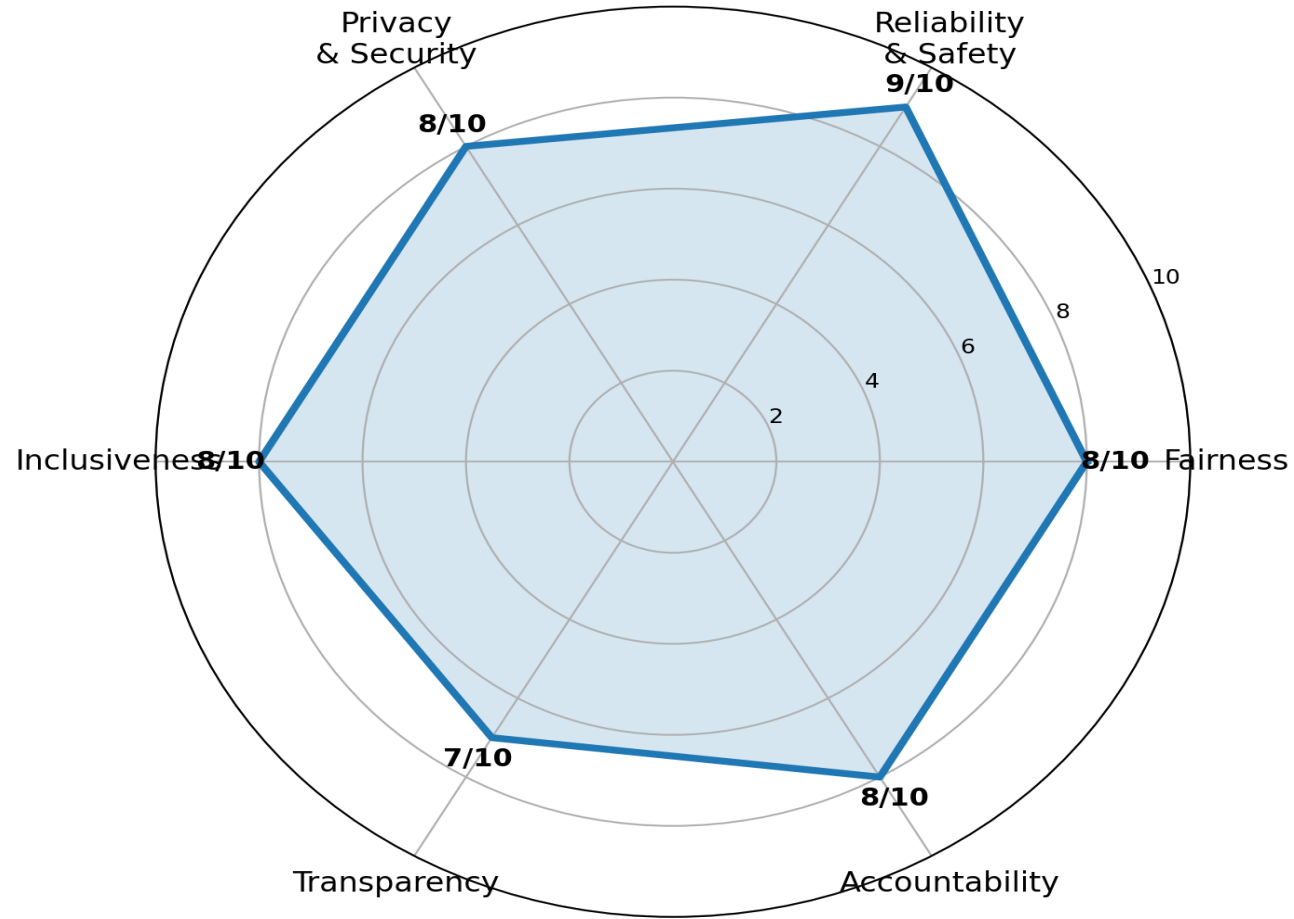


A police officer watches a television monitor displaying a fraction of London's CCTV camera network

Photo: Daniel Berehulak ([Getty](#))

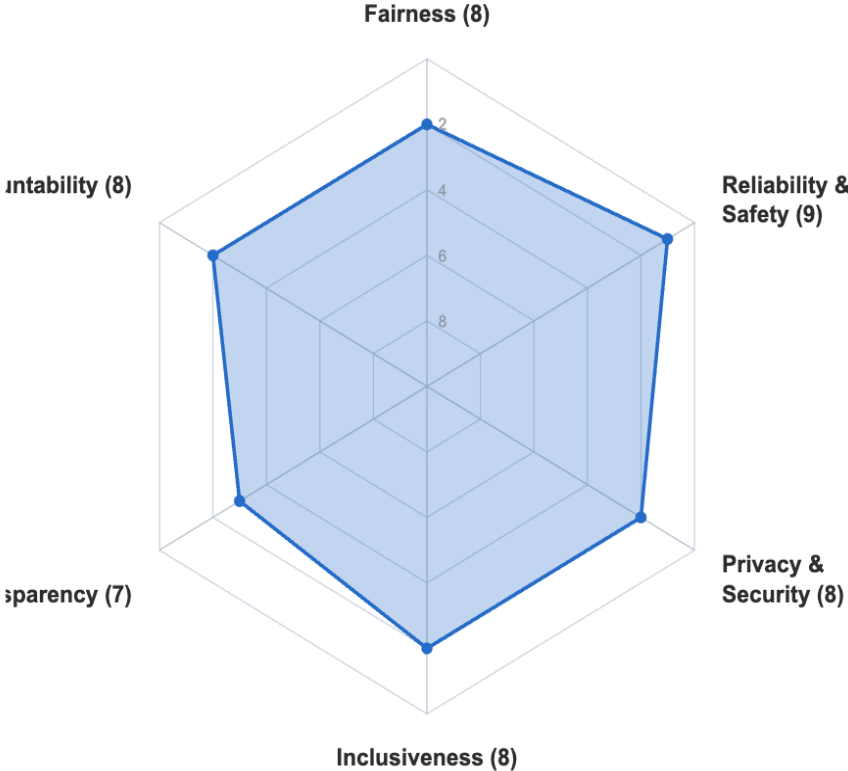
Anthropic Fable 5: Responsible AI Rating Using Microsoft AI Principles

Fable 5 Evaluation



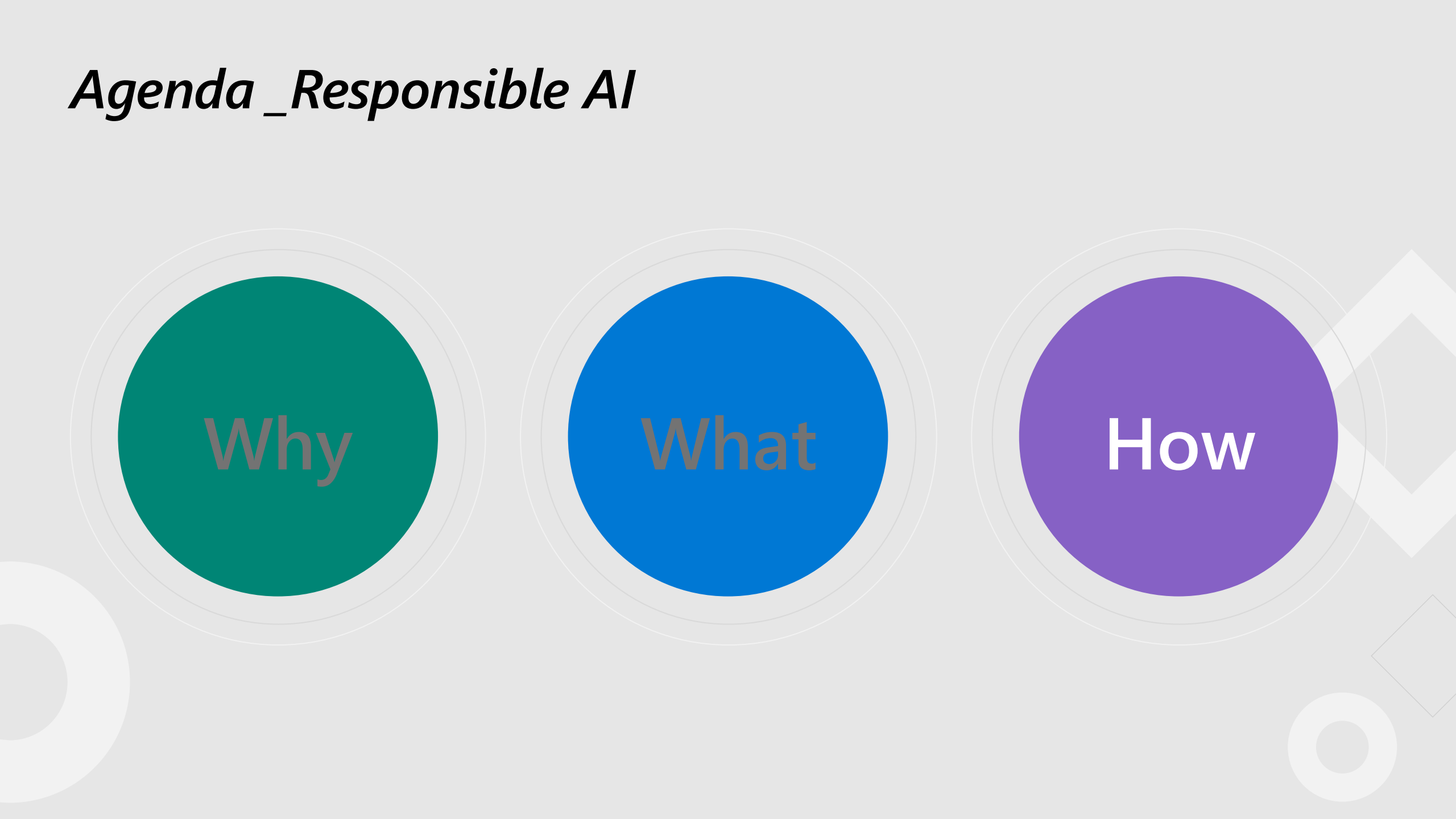
Interpretation: strongest in reliability & safety; improvement area: transparency.
Scale: 1 = weak alignment, 10 = strong alignment.

Fable 5 Evaluation



Principle	Rating	Comments
Fairness	8/10	Anthropic has invested heavily in Constitutional AI and bias mitigation. The model generally avoids discriminatory outputs and attempts balanced responses. However, like all LLMs, bias can still emerge from training data and prompting patterns.
Reliability & Safety	9/10	This is arguably Anthropic's strongest area. Claude models consistently score well on harmful content resistance, jailbreak robustness, and safe completion behavior. Extensive red-teaming and constitutional training contribute to strong safety performance.
Privacy & Security	8/10	Anthropic provides enterprise-grade security controls, data handling policies, and API protections. Stronger ratings depend on deployment architecture, customer-controlled retention settings, and audit capabilities.
Inclusiveness	8/10	Supports multiple languages and generally accommodates diverse user backgrounds and accessibility needs. There is still room for improvement in low-resource languages and cultural representation.
Transparency	7/10	Anthropic publishes substantial research, model cards, safety reports, and system descriptions. However, like most frontier model providers, training datasets, model weights, and many internal mechanisms remain proprietary.
Accountability	8/10	Anthropic has governance structures, safety teams, published responsible scaling policies, and documented risk management processes. Enterprise customers benefit from monitoring and policy controls. Full accountability remains challenging for any frontier AI provider.

Agenda _Responsible AI



Why

What

How

Putting Responsible AI into Practice



Responsible AI & Human Rights



Dignity of every individual



Freedom from discrimination



Freedom from invasions of privacy



Freedom of expression



Freedom of association



THE UNIVERSAL DECLARATION OF Human Rights



Article 1 — All human beings are born free and equal in dignity and rights. They are endowed with reason and conscience and should act towards one another in a spirit of brotherhood.

Article 2 — No one shall be subjected to discrimination of any kind, whether on the basis of race, colour, sex, language, religion, political or other opinion, national or social origin, property, birth or other status.

Article 3 — Everyone has the right to life, liberty and security of person.

Article 4 — No one shall be held in slavery or servitude; slavery or servitude shall be prohibited in all its forms.

Article 5 — No one shall be subjected to torture or to cruel, inhuman or degrading treatment or punishment.

Article 6 — Everyone has the right to a fair and public hearing by an independent and impartial tribunal.

Article 7 — All are equal before the law and are entitled without any discrimination to equal protection of the law. All are entitled to equal treatment before the courts.

Article 8 — Everyone has the right to an effective remedy by the competent national or international institutions.

Article 9 — No one shall be subjected to arbitrary arrest, detention or exile.

Article 10 — Everyone has the right to a fair and public hearing by an independent and impartial tribunal.

Article 11 — 1. Everyone has the right to a fair and public hearing by an independent and impartial tribunal. 2. The accused shall be presumed innocent until proven guilty. 3. No one shall be held guilty of any crime without a fair trial.

Article 12 — No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation. Everyone has the right to the protection of the law against such interference or attacks.

Article 13 — 1. Everyone has the right to free movement and residence within the borders of each state. 2. Everyone has the right to leave any country, including his own, and to return to his country.

Article 14 — 1. Everyone has the right to seek and to enjoy in other countries asylum from persecution. 2. No one shall be subjected to persecution.

Article 15 — 1. Everyone has the right to a nationality. 2. No one shall be arbitrarily deprived of his nationality nor denied the right to change his nationality.

Article 16 — 1. Men and women of full age, without any limitation of race, nationality or religion, have the right to marry and to found a family. They are entitled to equal rights in marriage. 2. Marriage shall be entered into only with the free and full consent of the intending spouses. 3. The family is the natural and fundamental group unit of society and is entitled to protection by society and the State.

Article 17 — 1. Everyone has the right to own property alone as well as in association with others. 2. No one shall be arbitrarily deprived of his property.

Article 18 — Everyone has the right to freedom of thought, conscience and religion; this right includes freedom to change his religion or belief.

Article 19 — Everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without restriction and to receive and impart information and ideas without any interference by public authorities.

Article 20 — 1. Everyone has the right to peaceful assembly and to associate with others in peaceable manner. 2. No one shall be compelled to belong to an association.

Article 21 — 1. Everyone has the right to take part in the government of his country, directly or through freely chosen representatives. 2. Everyone has the right of equal access to public service in his country. 3. The will of the people shall be the basis of the authority of government; this will shall be expressed through periodic and genuine elections which shall be by secret ballot and shall ensure the free expression of the will of the voters.

Article 22 — Everyone has the right to social security.

Article 23 — 1. Everyone has the right to work, to free choice of employment, to just and favourable conditions of work and to protection against unemployment. 2. Everyone, without any discrimination, has the right to equal pay for equal work. 3. Everyone has the right to a fair and favourable remuneration ensuring for himself and his family a standard of living adequate to the health and well-being of himself and his family, including housing, food, clothing, medical care and social services. 4. Everyone has the right to a fair and favourable remuneration ensuring for himself and his family a standard of living adequate to the health and well-being of himself and his family, including housing, food, clothing, medical care and social services.

Article 24 — Everyone has the right to rest and leisure, including reasonable limitation of working hours and periodic holidays with pay.

Article 25 — Everyone has the right to a standard of living adequate for the health and well-being of himself and his family, including food, clothing, housing, medical care and social services.

Article 26 — 1. Everyone has the right to education. 2. Education shall be free, at least in the elementary and fundamental stages. 3. Elementary education shall be compulsory. 4. Technical and vocational education shall be made generally accessible and higher education shall be open to all on the basis of merit.

Article 27 — 1. Everyone has the right to the free enjoyment of his own material and intellectual property. 2. No one shall be arbitrarily deprived of his property.

Article 28 — Everyone has the right to a social and international order in which the rights and freedoms set forth in this Declaration can be fully realized.

Article 29 — 1. Everyone has duties to the community in which alone the free and full development of his personality is possible. 2. These rights and freedoms shall be exercised by everyone in conformity with the laws of the country, and such laws shall be limited by the rights and freedoms of others.

Article 30 — No one shall be subjected to arbitrary arrest, detention or exile.

Microsoft Global Human Rights Statement

Information technology is transforming human experiences at an accelerating pace. As technology evolves, it touches lives in new ways. The statement explains how Microsoft is committed to respecting and promoting human rights, to ensure that technology plays a positive role across the globe.

Putting responsible AI into practice

Principles

Fairness
Accountability
Transparency
Inclusiveness
Reliability & Safety
Privacy & Security

Putting responsible AI into practice



Human-AI Guidelines

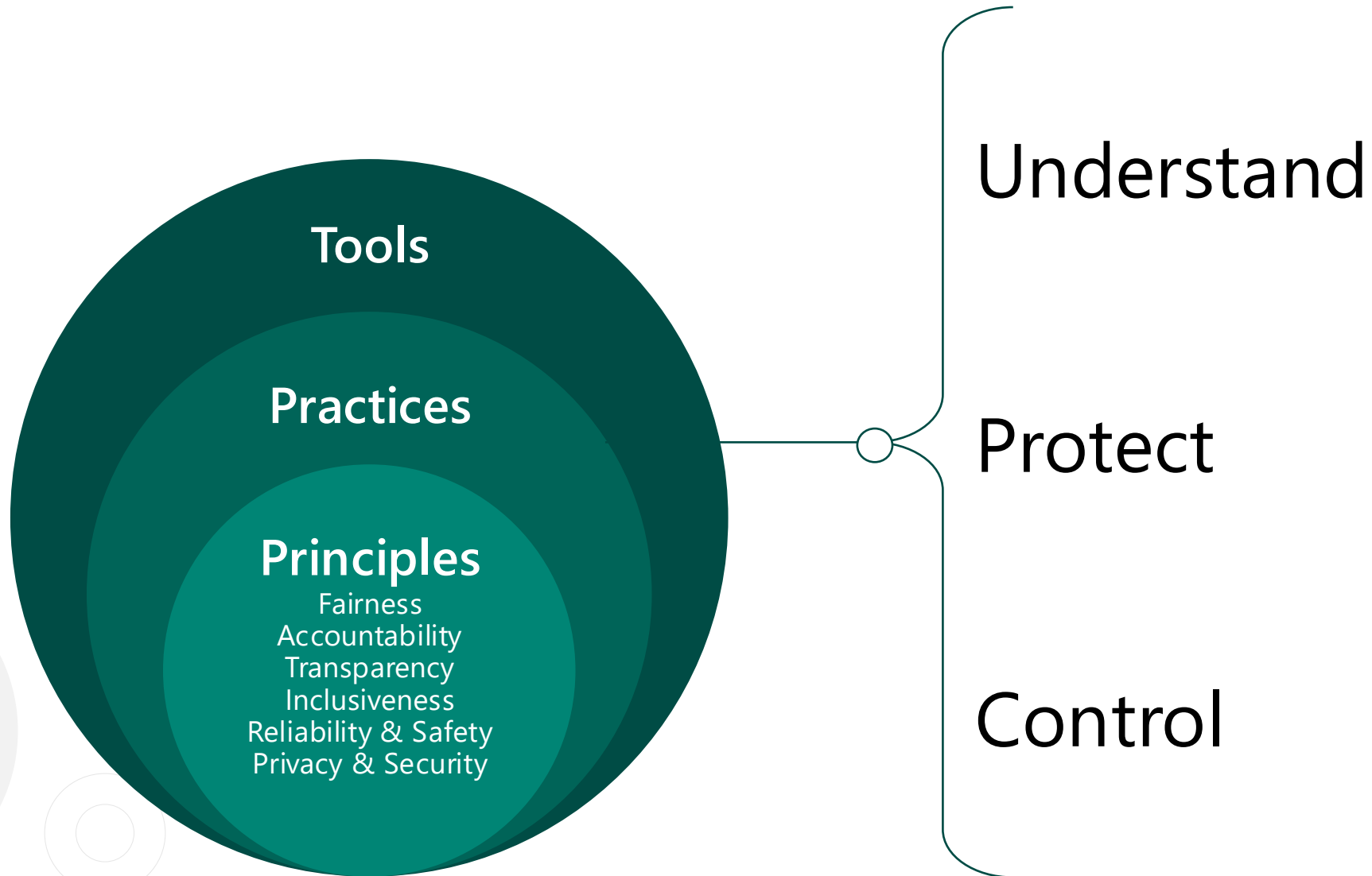
Conversational AI Guidelines

Inclusive Design Guidelines

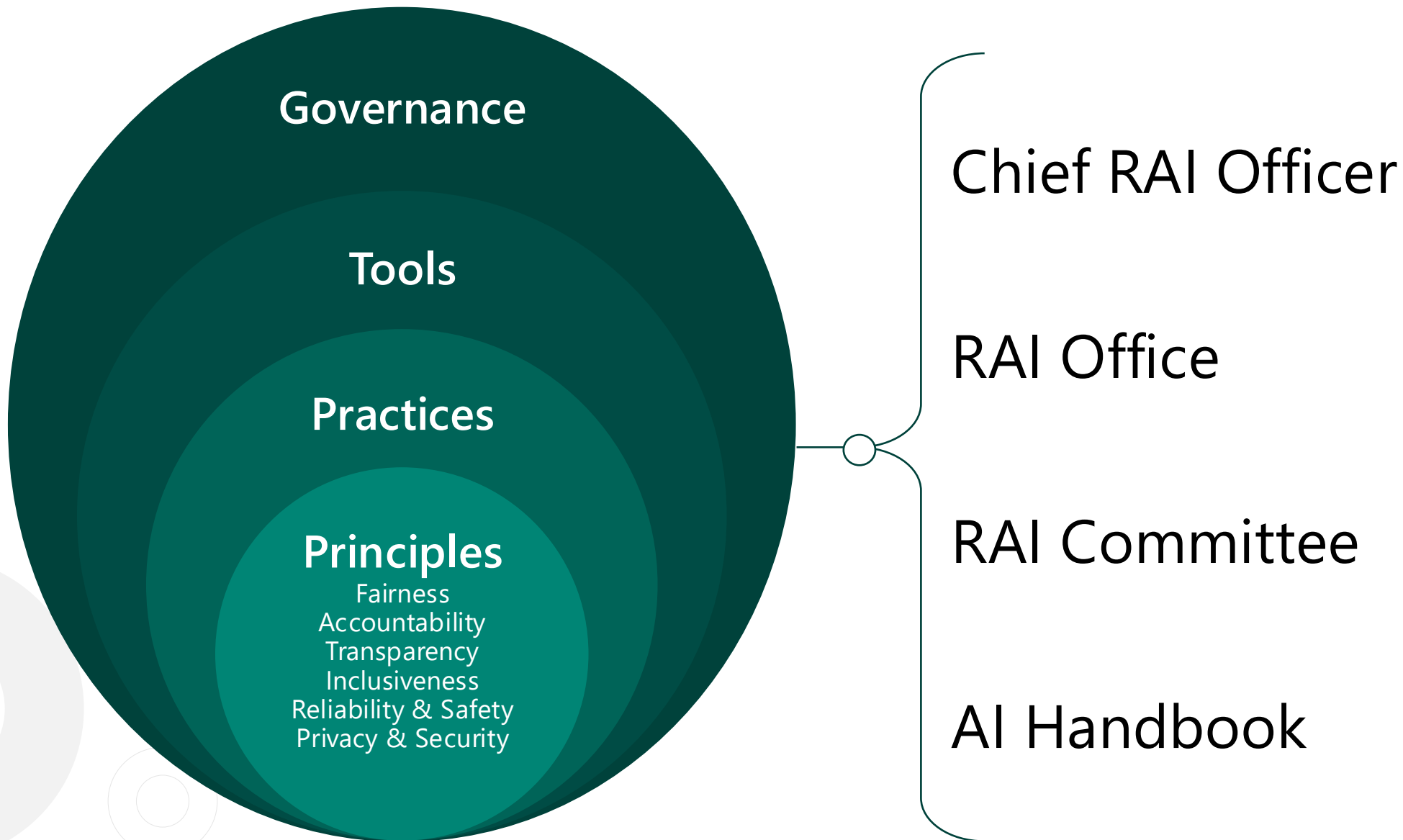
AI Fairness Checklist

Datasheets for Datasets

Putting responsible AI into practice



Putting responsible AI into practice



IBM Ethical AI

Principles for Trust and Transparency

- 1 The purpose of AI is to augment — not replace — human intelligence
- 2 Data and insights belong to their creator
- 3 New technology, including AI systems, must be transparent and explainable

Pillars of Trust



Explainability

An AI system's ability to provide a human-interpretable explanation for its predictions and insights



Fairness

Equitable treatment of individuals or groups by an AI system — depends on the context in which the AI system is used



Robustness

An AI system's ability to effectively handle exceptional conditions, such as abnormalities in input



Transparency

An AI system's ability to include and share information on how it has been designed and developed



Privacy

An AI system's ability to prioritize and safeguard consumers' privacy and data rights

Responsible AI by Design

- ✓ AI is embedded in everyday life, business, government, medicine and more.
- ✓ **You** will be helping people and organizations adopt AI responsibly.
- ✓ Only by embedding ethical principles into AI applications and processes can we build systems based on trust.

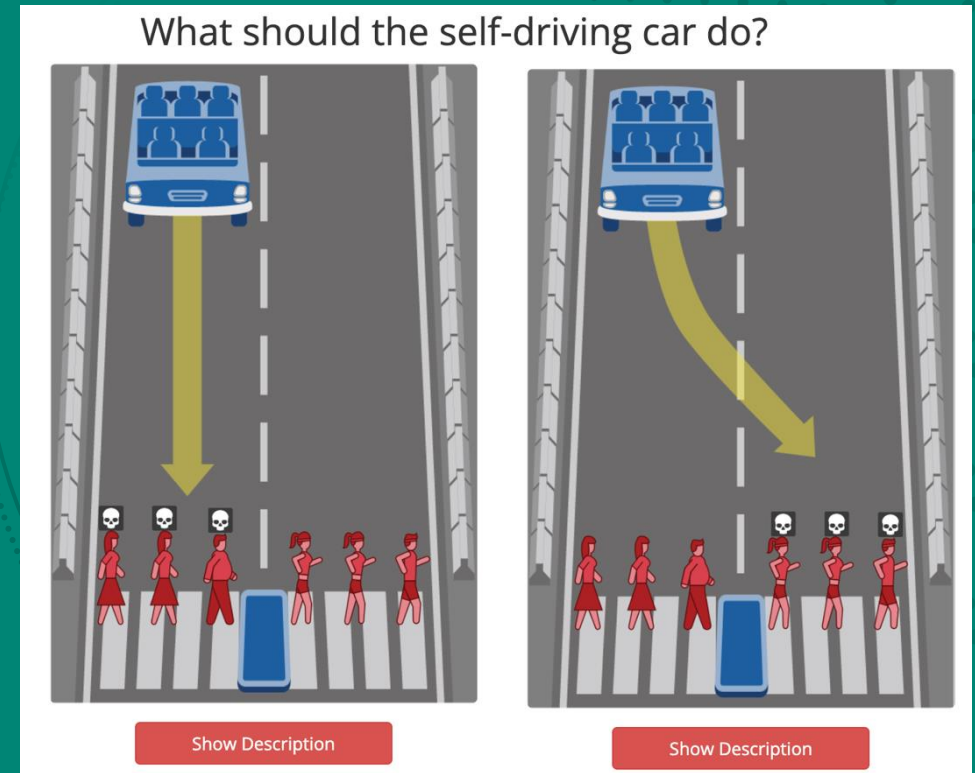


Q&A



Lab

www.moralmachine.net/



<https://www.colorado.edu/project/imagine-ai/moral-machine>